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PLOS One

RESEARCH ARTICLE

Perennial plants and traditional medicine: Assessing the socio-demographic impacts on traditional remedies

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Citation: Wang X, Zheng Y, Cui X, Anwar T, Qureshi H (2025) Perennial plants and traditional medicine: Assessing the socio-demographic impacts on traditional remedies. PLoS One 20(8): e0328913. https://doi.org/10.1371/journal.pone.0328913

Editor: Asghar Khan, Government Degree College Totakan, PAKISTAN

Received: October 2, 2024

Accepted: July 4, 2025

Published: August 8, 2025

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Data availability statement: The author confirms that all data generated or analyzed during this study are included in this published article.

Funding: This work was funded by the following: 1. Research on the Application oriented Characteristic Construction of Undergraduate Colleges in Jilin Province from the Perspective

Abstract

This study assessed the curative uses of wild medicinal plants and investigated the socio-demographic aspects influencing traditional medicinal knowledge in Tehsil Hasilpur, Punjab. Using snowball sampling, data were collected from 300 respondents through interviews and field surveys. Analysis included plant life span, plant parts used, preparation methods, and quantitative indices such as Informant Consensus Factor (ICF), Use Value (UV), and Fidelity Level (FL). The findings revealed that males comprised 85.6% of respondents, reflecting cultural constraints on women's participation. Traditional knowledge was more prevalent among individuals with primary education (53%) and the illiterate (24%). Perennial species dominated (98.6%), leaves are the most commonly utilized plant part (65.33%), and decoctions are the most popular preparation method (24%). High ICF values indicated strong consensus on plant efficacy, with *Eucalyptus globulus* exhibiting significant use value. These findings emphasize the impact of gender, education, and occupation on knowledge transfer, as well as the ecological significance of local flora. The study recommends integrating traditional medicinal knowledge into healthcare strategies and promoting collaborative conservation efforts to ensure sustainable use of ethnomedicinal resources.

Introduction

Ethnobotany explores the intricate relationship between human culture, biodiversity, and traditional ecological knowledge, rooted in centuries of indigenous wisdom and scientific inquiry. It examines the diverse ways different cultures interact with plants, contributing to both scientific understanding and the preservation of cultural heritage [1]. As a multidisciplinary field, ethnobotany serves as a repository for traditional medicinal practices and plays a vital role in biodiversity conservation, sustainable resource management, and the discovery of novel therapeutic compounds [2]. By

of Education Ecology in the Digital Age. — This work was sponsored in part by Jilin Higher Education Association (JGJX2023D839). 2. Research on the Application of Stable Diffusion in Landscape Architecture Design in the AI Era. — This work was sponsored in part by Changchun SCI-TECH University. The recipients of the above funding are Xiaonan Wang, Yi Zheng, and Xin Cui. The funders had no role in study design, data collection and analysis, decision to publish, or preparation of the manuscript.

Competing interests: The authors have declared that no competing interests exist.

systematically documenting and analyzing ethnobotanical knowledge, researchers uncover valuable insights into the uses, properties, and conservation status of plants, particularly their therapeutic potential. This includes exploring their antimicrobial, antioxidant, antiviral, anticancer, anti-inflammatory and sedative properties, which help identify bioactive compounds for developing effective pharmaceuticals or integrating them into traditional and complementary medicine practices [1,3]

Ethnomedicine, which focuses the study of traditional healing techniques among indigenous and ethnic people, has dual aim: it preserves traditional knowledge and facilitates the development of innovative treatments [4]. Due to its significant contribution to the diversity and formulation of modern medicine, ethnomedical knowledge is globally recognized and valued [5]. As traditional plant-based healing practices face the risk of being lost due to variables such as information deterioration, biodiversity loss, and socioeconomic shifts, it becomes crucial to thoroughly document and study this knowledge across different cultural groups [6]. Despite the advances in modern science and medical technologies, many communities still depend heavily on indigenous plant knowledge for remedies [7]. The World Health Organization (WHO) defines traditional medicine as a collection of knowledge, skills, and practices developed from different cultural views and human experiences [8].

In isolated regions of poor nations with little access to modern healthcare, nearly 80% of the population depends on traditional medicine, which is often produced from medicinal plants [9]. These plants not only hold therapeutic potential but also serve as valuable commodities in urban markets, significantly influencing the livelihoods of local communities [10]. Among Native American communities, Medicinal plants have traditionally played an important role in healing human and animal diseases. As emphasized by [11], indigenous knowledge has been vital to scientific progress, particularly in the development of plant-based remedies for modern pharmaceuticals. In fact, over 25% of contemporary pharmaceuticals are derived from natural sources, many of which have roots in traditional use. Unfortunately, the decline in the oral transmission of home remedies across generations is leading to a reduction in both the use of traditional medicinal herbs and the knowledge surrounding them [12]. Prior to the introduction of insulin and other blood glucose-lowering drugs, traditional herbal treatments were widely utilized to treat diabetes and its consequences. Research by [13] has identified over 1,200 medicinal plants with potential antidiabetic properties. Comparative studies on therapeutic plants, whether across different nations, ethnic groups, or regions, help identify the most effective species for treating various health issues, such as liver-related conditions [14].

Local herbalists play an important role in addressing common health issues such as respiratory ailments, musculoskeletal disorders, aches, pains, and wounds. In many communities, individuals often rely on medicinal herbs based on personal experience, sometimes bypassing consultations with traditional healers [15]. Herbal remedies remain an integral part of healthcare practices across various cultures worldwide. Ethnobotanical surveys and ethno-pharmacological research are Essential instruments for recognizing and documenting medicinal plants [16]. For example,

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Cannabis sativa extract is valued for its medicinal properties, including anti-emetic, anti-epileptic, anti-parasitic, diuretic, anti-inflammatory, anti-bacterial, antipyretic, anti-tumor, vermifuge, carminative, vermicide, and pain-relieving effects, and has a rich history of use in Arabic-Islamic medicine [17]. However, factors such as agricultural development, modernization, population growth, and climate change are contributing to The loss of traditional knowledge, which is having a profound impact on local communities. Urbanization further accelerates this decline as younger generations migrate from rural areas in search of employment opportunities and urban lifestyles [18].

Pakistan, with its rich biodiversity and varied climate zones, is a key source of medicinal herbs, where rural communities rely on agriculture, livestock, and herbal collection for income. However, overexploitation and lack of preservation efforts threaten ethnomedicinal plants, calling for stronger conservation measures. Collaboration between educational institutions, agricultural departments, and local authorities is crucial for sustainable plant cultivation and improving community livelihoods. Despite the abundance of medicinal plant knowledge, the lack of comprehensive data emphasizes the need for systematic documentation. Ethnomedicine in Pakistan, with over 5,700 medicinal plant species, depends on traditional knowledge for healthcare, yet suffers from poor documentation [19]. This study in Tehsil Hasilpur aims to document medicinal plants, uncover previously undocumented species, and provide insights into their traditional uses, contributing to the conservation of botanical resources and cultural heritage. Key objectives include identifying commonly used plants, comparing findings with existing literature, and promoting sustainable plant management.

Materials and methods

Study area

The Tehsil Hasilpur is in the Bahawalpur District of Punjab, Pakistan georeferenced at approximately 29.7129° N latitude and 72.5540° E longitude (Fig 1).

The study area is characterized by various forest types including mixed deciduous forests and riverine forests, with dominant vegetation such as *Acacia* species, *Olea europaea* and *Eucalyptus globulus*. The climate is arid, with hot summers and mild winters.

Health facilities in the area include government hospitals, clinics, and dispensaries catering to the healthcare needs of the population. The education status of the research area shows a mix of educational institutions., ranging from primary schools to higher secondary schools. Traditional and cultural norms play a significant role with local communities often relying on indigenous knowledge for healthcare practices and utilizing traditional medicines from plants like *Senegalia catechu* and *Ficus religiosa* (Fig 2). These cultural practices are deeply rooted and contribute to the unique socio-cultural fabric of the region [20].

Survey of the area and data collection

Several pre-data collection surveys were conducted to identify suitable collection sites and to become familiar with the area. The sampling and interviews took place between 15th March and 20th September 2023. The first research purpose was to collect and identify wild plants in Hasilpur strictly adhere to the ISE code of ethics (http://ethnobiology.net/code-of-ethics/) during data gathering. Direct interviews were conducted using a structured questionnaire, supplemented by the snowball sampling technique in which initial participants help identify additional respondents within their social networks. We introduced ourselves as researchers and established a brief relationship with respondents to gain their trust before beginning interviews. Participants provided verbal consent in front of a witness to guarantee full compliance with ethical norms. Given that most interviewees, such as farmers, housewives, and hakims (local practitioners), For those who did not speak English, a questionnaire was translated into their original language (Punjabi), and their responses were recorded.

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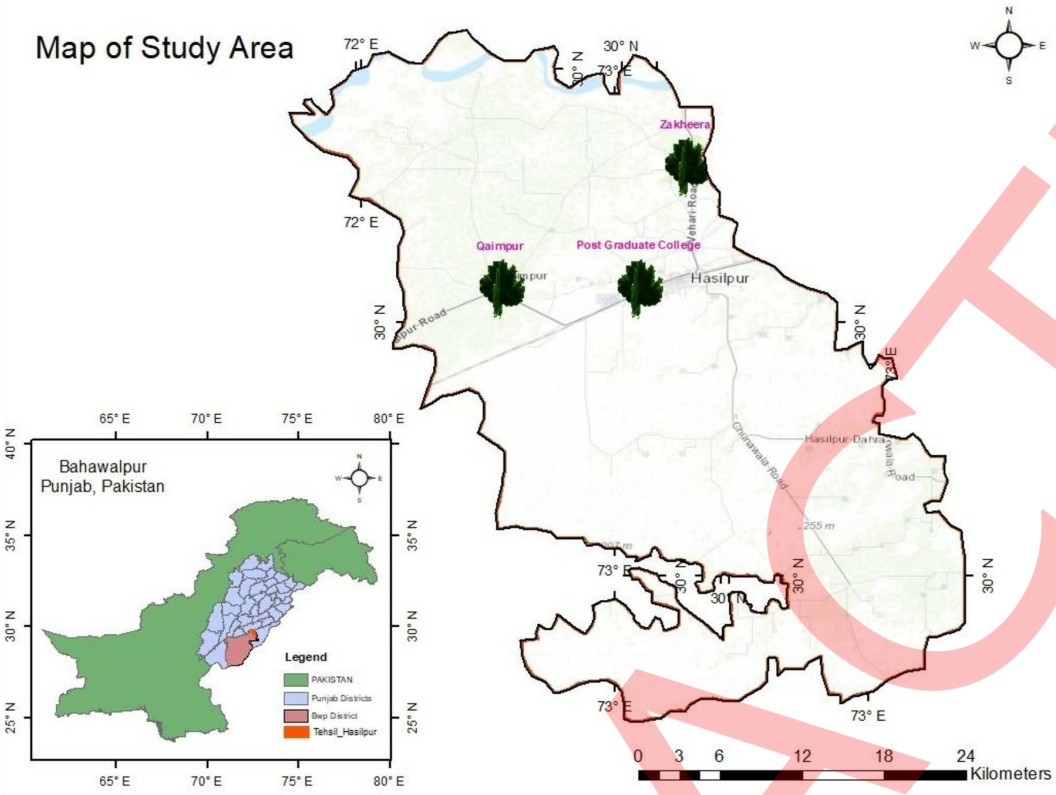


Fig 1. Map of the study area, generated using ArcGIS software (Esri, Redlands, CA, USA).

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Locals contributed information on local names, recipes, local usage, plant parts used, and participation in data collecting. The local names and growth characteristics were recorded for each plant species with curable properties. After each interview, sample specimens are systematically numbered, pressed, and dried was conducted to facilitate ultimate identification, utilizing the Pakistan e-flora (http://www.tropicos.org/Project/Pakistan) as a valuable reference source. The obtained plant samples were thoroughly processed, including shade drying, pressing, categorization, and identification by Dr. Ghulam Sarwar, Botany Department, The Islamia University of Bahawalpur. The herbarium records were carefully labeled and kept for future reference (Fig 3). Our experiment adheres to the applicable institutional, national, and international rules and legislation.

Quantitative ethnomedicinal analysis

Qualitative and quantitative analysis methods were used to evaluate the data. Microsoft Excel v. 2019 and SPSS v. 25 were used for this purpose.

Informant Consensus Factor (ICF)

ICF was used to validate agreement between participants and plants over a certain sickness category. It was calculated using the specified formula [21].

$$ICF = \frac{Nur - Nt}{(Nt - 1)}$$

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Fig 2. Sceneries of the study area taken by the authors during field visits (a) General landscape view of Tehsil Hasilpur, highlighting the semi-arid environment (b) Wild medicinal flora growing in natural habitats (c) Perennial plant species dominant in the area (d) Field survey with a local informant sharing ethnomedicinal knowledge during data collection (e) A typical household garden where some medicinal plants are cultivated for domestic use (f) Rural household surroundings.

https://doi.org/10.1371/journal.pone.0328913.g002

Nur signifies the use reports for a given illness group, and Nt denotes the total number of species used for that disease group. Its value ranges between 0 and 1. The high value (approaching 1) shows that plant species are picked according to specific characteristics, and knowledge among participants is highly communicated.

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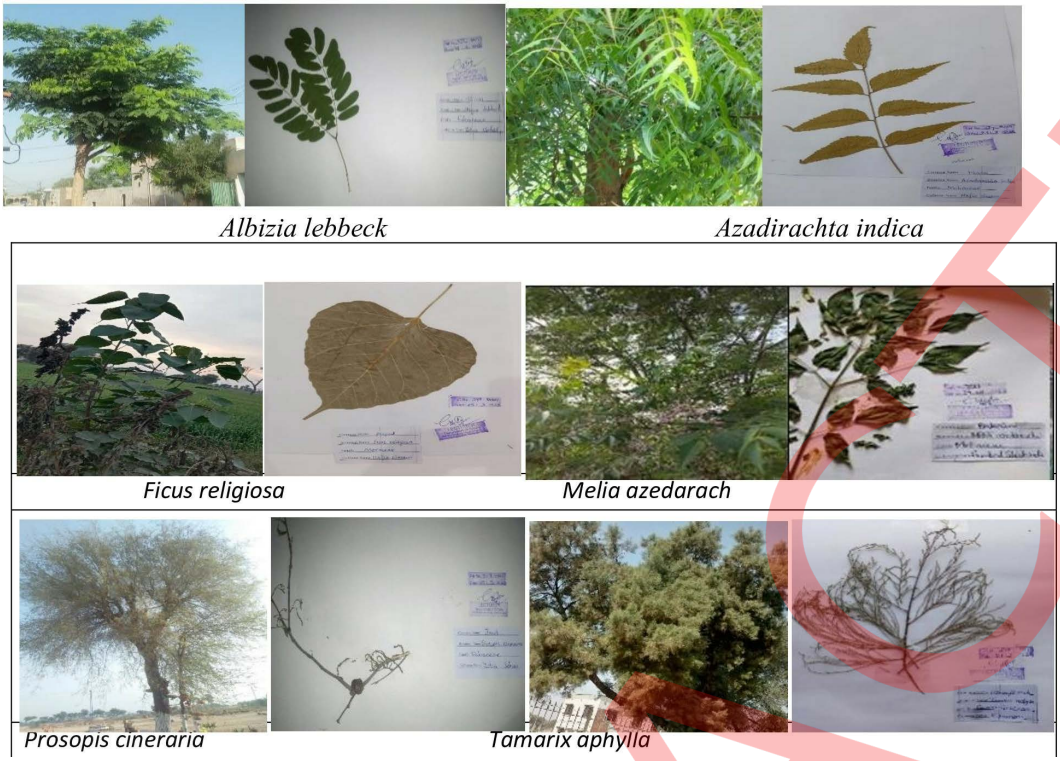


Fig 3. Ethnomedicinal significant plants of the study area.

https://doi.org/10.1371/journal.pone.0328913.g003

Use Value (UV)

It is a quantitative measure of the relative relevance of locally recognized plant species and was determined by applying the following formula [21].

$$UV = \frac{\sum Ui}{N}$$

Ui is the total uses cited by each participant for a specific species, and N is the total number of participants.

Fidelity Level (FL)

Ui is the total uses cited by each participant for a specific species, and N is the total number of participants [21].

$$FL\,(\%) = \frac{Np}{N} \times 100$$

Whereas "Np" shows the participant number that cites plant use for treating specific diseases and "N" is the total number of participants citing plant species for any ailment.

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Comparison of ethnomedicinal data Jaccard Index (JI)

The data given in this study was compared to previously published literature from adjacent regions. The work's uniqueness was determined by Comparing the presented statistics with previously published research at the regional and global levels. It was determined using the provided formula [21].

$$JI = \frac{c \times 100}{(a + b) - c}$$

Where "a" shows species of the study area, "b" shows species of the adjoining area and c shows mutual plant species in both areas.

Results and discussion

Tehsil Hasilpur is renowned for its diverse array of medicinal plants, prompting a preliminary survey aimed at collecting data on ethnomedicinally significant plants and their folk medicinal applications. Through a combination of verbal communication and structured questionnaires, residents in the research region were interviewed to discover the utilization of these plants in treating various ailments. The ecological conditions of the site play a vital role in fostering the abundance of medicinal wild plants. The findings of this study emphasize the importance of herbal medicinal systems in the broader context of advancing the development of future medications. By conducting scientific investigations, there exists an opportunity to reinforce our capabilities in addressing a wide spectrum of ailments, particularly in environments where contemporary medications may be inaccessible or susceptible to rapid degradation.

Socio-demographic data of informants

In a demographic analysis of 300 respondents concerning therapeutic wild plants, a significant gender gap emerged, with males comprising 85.6% and females only 14.3%, potentially reflecting cultural norms or societal divisions in tasks and responsibilities, where men might have greater access to outdoor environments where wild plants grow or be more involved in activities related to plant gathering. Examination of educational backgrounds revealed that traditional knowledge of therapeutic herbal use was prevalent among those with primary education (53%) and the illiterate (24%), suggesting that such knowledge might be passed down through generations or acquired through informal channels within communities, bypassing the need for formal education. Occupationally, the village population exhibited diverse roles, with farmers constituting 49% and other occupations, including housewives and herbalists, collectively making up 40% of respondents. Farmers and housewives demonstrated a deeper understanding of plant uses, possibly due to their close connection to traditional lifestyles and reliance on natural remedies for healthcare needs, emphasizing the influence of occupational engagement and daily interactions with the environment on knowledge acquisition. Furthermore, the reliance on medicinal plants among low-wage workers, compounded by limited access to allopathic medicines, shows the socioeconomic disparities in healthcare access. These findings highlight the intricate interplay of socio-cultural dynamics in shaping knowledge dissemination and conservation efforts surrounding traditional healthcare practices, emphasizing the need for holistic approaches that consider gender roles, educational backgrounds, and occupational engagement to promote sustainable healthcare practices rooted in local knowledge systems.

Categorizing participants by age into four groups—30–40 age group (28%), 41–50 age group (53%), 51–60 age group (13%), and above 60 years (3%)—revealed a notable awareness of herbal use within the 41–50 age group. This distribution suggests that older individuals possess more significant knowledge about medicinal plants compared to middle-aged and young individuals, as they are better connected to traditional lifestyles and have accumulated extensive experience with medicinal plant use over time younger people, on the other hand, are more likely to abandon traditional herbal practices due to changing lifestyles and the impact of industrialization, which may lead to a decline in traditional knowledge

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transmission across generations (Table 1). This disparity highlights the importance of intergenerational knowledge transfer efforts to preserve and promote traditional healthcare practices in the face of sociocultural shifts and modernization.

Aligned with findings from other research studies, [22] reported similar results in their investigations conducted in the Chenab Riverine Area, Punjab. Typically, essential information concerning medicinal plants is passed down orally through generations rather than being documented in written form, creating a looming risk that this essential knowledge will dissipate in the future due to a lack of interest among younger folks. The majority of interviewed male respondents were primarily from the 41–60 age group. Additionally, the surveyed residents were predominantly either illiterate or had received only primary education, with farming being their predominant occupation [23].

Plant species life span

The vegetation in Tehsil Hasilpur predominantly comprises perennial trees, with a lesser prevalence of biennials, and notably, perennials constitute a substantial majority, accounting for 98.6% of the plant species. This prevalence can be linked to the hard desert climate of the region, forcing plants to adopt methods to endure severe conditions and optimize seed production during favorable periods. The dominance of perennials aligns with the region's climate, where the vegetation has evolved to cope with water scarcity and extreme temperatures. Perennials exhibit adaptability to challenging environments by enduring harsh conditions and developing strategies for efficient seed production during favorable periods. Moreover, the prevalence of herbs and shrubs further underscores their resilience and ability to thrive in short-growing cycles, rapid flowering, and effective seed production. These findings highlight the adaptive strategies of the local flora to the specific ecological challenges posed by Tehsil Hasilpur's arid climate, offering valuable insights for conservation efforts and sustainable management practices in the region. Additionally, it contributes to the broader understanding of how plant communities respond to environmental stressors, which is essential for devising conservation and land management strategies in arid regions. This aligns with findings from Dunyapur, District Lodhran, where the majority of plants exhibit a perennial life span, with annual and biennial categories following in frequency [3].

Family contribution to medicinal flora

During the survey, a comprehensive analysis was conducted on selected medicinal trees. Many of these plants exhibited favorable xerophytic characteristics, showing their remarkable ability to endure prolonged periods of drought. The

Table 1. Socio-demographic data of informants.

Variation	Category	Number	Percentage
Gender	Male	257	85%
	Female	43	14%
Age	30-40	84	28%
	41-50	159	53%
	51-60	41	13%
	Above 60	9	3%
Occupation	Farmer	148	49%
	Pansar	8	2%
	Herbalist	22	7%
	Others	122	40%
Educational Background	Illiterate	74	24%
	Primary	159	53%
	Middle	37	12%
	Matric	33	11%
	Above Matric	8	2%

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study area showed a rich plant diversity, encompassing a total of 14 different families. According to the survey findings, Fabaceae stands out as the largest family, constituting 20% of the total, with four distinct species. Following closely are Moraceae, Tamaricaceae, and Oleaceae, each contributing 10% to the overall family composition as depicted in Fig 4. The survey outcomes shed light on the ecological adaptability of medicinal trees in Tehsil Hasilpur, emphasizing their prevalence in arid conditions and resilience to prolonged droughts. The dominance of xerophytic characteristics among the studied plants emphasizes their ability to thrive in challenging climatic conditions.

Haripur district documented medicinal plant species affiliated with prominent plant families such as Lamiaceae, Moraceae, Apocynaceae, Asteraceae, and others. The dissemination of traditional knowledge related to plant families has been widely reported globally [24]. Similarly, at Multan, identified dominant plant groups including Asteraceae, Fabaceae, and Amaranthaceae, aligning with the outcomes of our present study. Another study conducted in the Hangu District highlighted dominating plant groups, particularly Asteraceae and Solanaceae [25]. Interestingly, only seven species are similar between our current study and the one reported from District Bahawalpur, despite the geographical proximity within the same division of the Punjab province. This discrepancy arises because respondents in the previously published literature were not confined to reporting solely wild plants; they also included cultivated herbs or shrubs alongside trees. In District Haripur has recorded the use of 80 plant species from 50 families as therapeutic herbs, with Lamiaceae being the predominant family with seven species. Additionally, 40 plant species from 39 genera and 32 families in the Haripur region of Pakistan, emphasize the therapeutic potential of Amaranthaceae in treating conditions such as skin diseases, intestinal worms, and inflammations [8].

Plant part used in ethnomedicinal remedies

The observed application of diverse plant parts for the treatment of human ailments in the studied area demonstrates a strong dependence on traditional medical practices among the locals. Notably, leaves constituted the major portion of medicinal plant collection, accounting for 65.33% of the remedies, while fruits contributed to 34.4%. The preference for leaves can be attributed to their abundance, which contains a diverse array of chemical compounds with therapeutic properties. Their ease of extraction and versatility in medicinal formulations make them a preferred resource for addressing

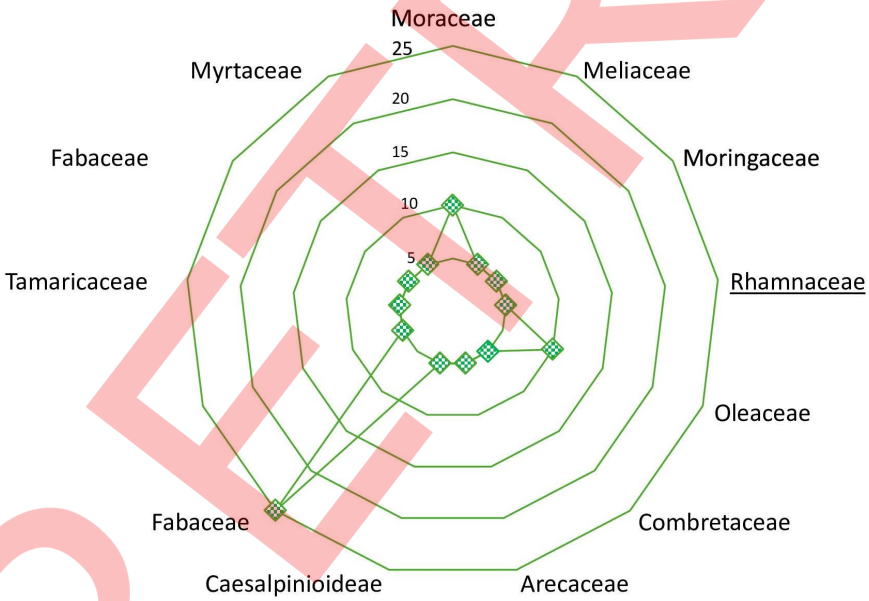


Fig 4. Indigenous medicinal plants' diversity.

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various health issues in the community. The focus on plant parts in this study provides useful insights into local knowledge and practices related to medicinal flora, contributing to the ongoing discourse on ethnobotanical research in the region. Consistent findings across various studies highlight the prevalence of leaves as the predominant plant parts used in herbal preparations. Research conducted in Khyber Pakhtunkhwa (KPK) also identified leaves as the most widely used plant parts in traditional medical procedures, with fruits, stems, and roots following in usage [26]. A similar pattern was reported in the Chenab Riverine Area, Punjab, where leaves ranked highest in local preferences for crafting medicines, succeeded by roots, stems, fruits, and flowers. These results align with the observations of research conducted in the Hangu district, emphasizing the widespread utilization of leaves by traditional healers, followed by whole plants and fruits, in the preparation of medicinal remedies [26].

Methods of preparation of ethnomedicinal remedies

The results reveal the diverse methods of preparation utilized for medicinal plant consumption among respondents in the study area (Fig 5).

Decoction has emerged as the most popular approach, accounting for 24% of reported procedures, followed closely using juice at 18%. Powdered forms were less common, representing 7% of the reported methods, while extraction methods accounted for 8%. Surprisingly, the use of tablets was relatively low, constituting only 4% of reported practices. This diversity in preparation methods shows the adaptability and flexibility of traditional medicine practices, allowing individuals to tailor treatments according to personal preferences, cultural practices, and availability of resources. The prominence of decoction and juice preparation methods reflects a preference for liquid forms of administration, which may facilitate easier consumption and absorption of medicinal properties. Conversely, the relatively lower utilization of tablets suggests potential barriers to access or cultural preferences favoring more traditional preparation methods. Overall, these findings emphasize the need of evaluating varied preparation methods in the study region and the application of medicinal plants, acknowledging the diversity of traditional medicinal practices and their significance in healthcare provision within the community. Another study reported various traditional recipes for respiratory disease treatment involve diverse preparation methods, frequently impacted by life forms and specific plant parts used. Common modes of preparation include decoction (71.4%), extract (66.7%), infusion, paste (38.1%), and powder (33.3%). Less frequently employed methods encompass juice and ash (14.3%) [27]. Traditional healers typically combine two or more plant combinations with other substances such as water, milk, essential oils, honey, and butter. Decoction emerges as the most prevalent formulation, likely due to its practicality and effectiveness.

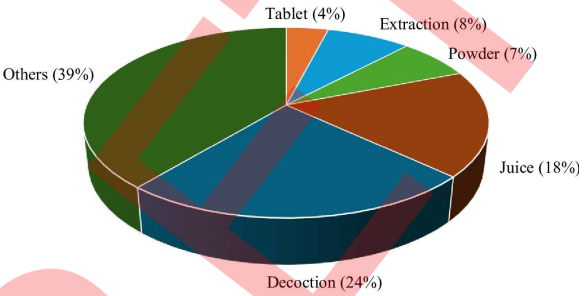


Fig 5. Methods of preparation of ethnomedicinal remedies.

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Quantitative analysis

The primary data collected during the survey was quantitatively assessed using the indices given below to compute homogeneity and validate the results.

Informant consensus factor (ICF)

The Informant Consensus Factor (ICF) is a quantitative measure used in ethnobotanical research to assess the degree of agreement among respondents regarding the use of specific plant species for treating particular health conditions. Higher ICF values indicate a stronger consensus, suggesting that certain plants are widely recognized within the community for their medicinal effectiveness. In this study, respiratory disorders, anti-inflammatory conditions, and bladder and kidney stones showed the highest ICF value (0.97), reflecting strong agreement among participants about the plants used to treat these ailments (Table 2). This high value underscores the collective confidence in the efficacy of these traditional remedies. Similarly, skin diseases, cough, cold, fever, and asthma exhibited ICF values of 0.77, 0.77, 0.87, and 0.78, respectively—demonstrating considerable but slightly lower consensus. In contrast, diuretics had a lower ICF value of 0.72, indicating a more varied response among informants about the plants used for this purpose. These ICF values provide meaningful insights into the strength and consistency of traditional medicinal knowledge in the community.

It ranges from 0 to 1, with higher values indicating stronger consensus and suggesting a more established or widely accepted use of particular plant species in treating that disease.

In contrast to findings in Hafiz Abad, where gastrointestinal disorders exhibited the highest ICF value of 0.39 [27], our study in Tehsil Hasilpur reveals that cholesterol-related issues have the highest ICF value. Similarly, a comparison with the Okara hepatitis study indicates that ulcers of the stomach and cancer held the highest ICF values (1.00) [28]. In a study performed in District Toba Tek Singh, Punjab, ICF values for swelling, snakebite, jaundice, liver disease, ulcer, and cough were reported as 0.45, 0.42, 0.28, 0.52, 0.60, and 0.42, respectively [1]. Notably, these values were considerably lower than those observed in our current study. Another study in District Lahore reported ICF ranges ranging from 0 to 0.45 for 21 species, closely resembling the values found in our research. While gastrointestinal illnesses displayed the greatest ICF value in the cited study, our research identified cholesterol as having the highest ICF value. The variation in ICF values across studies may be attributed to variations in local health concerns, traditional knowledge, and plant diversity. Studies in North-eastern Morocco [29], Jammu and Kashmir [30], and Chitral Valley [31] reported ICF values for diseases like skin diseases, diabetes, cancer, diarrhea, stomach disease, and gastrointestinal disorders. These values varied when compared with our study, showing the regional diversity in medicinal plant use and health priorities.

Table 2. Informant consensus factor (ICF).

Diseases	ICF
Skin diseases	0.53
Anti-inflammatory	0.568
Stomach Diseases	0.688
Wound Healing	0.822
Liver diseases	0.84
Diarrhea	0.913
Teeth treatment	0.953
Fever	0.966
Laxative	0.98
Anticancer	0.986

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Use value (UV)

The UV, ranging from 0.6 to 0.86, serves as a quantitative indicator of a species' significance as reported by the native inhabitants. *Eucalyptus globulus* attains the highest UV of 0.86, indicating its predominant use among the local population. Other notable plant species, with a UV of 0.6, include *Azadirachta indica*, *Melia azedarach*, *Moringa oleifera*, *Olea europaea*, *Prosopis cineraria*, *Ficus religiosa*, *Fraxinus ornus*, *Acacia catechu*, *Laurus nobilis*, *Dalbergia sissoo*, *Morus alba*, *Tamarix aphylla* and *Terminalia arjuna*. The use value for *Albizia lebbeck*, *Butea monosperma*, *Vachellia nilotica,* and *Ziziphus jujuba* is higher at 0.8 (Table 3). The utilization of quantitative value indices in this study provides valuable insights to investigate the relative importance of regionally identified plant species within ethnomedicinal practices. The UV values, ranging from 0.6 to 0.86, act as quantitative indicators, shining light on the significance of numerous species as stated by the indigenous people. *Eucalyptus globulus* emerges with the highest UV value of 0.86, underscoring its prevalent use among the local population. These quantitative metrics provide a full insight of the local preferences and practices related to various plant species aiding in the conservation and sustainable use of ethnomedicinal resources. In comparison with the findings from Hafiz Abad, *Solanum surattense* holds the highest UV value (0.86). A study in Morocco [32] reported the medicinal uses of 23 plant species, with only one plant matching the current study. The use values for *Olea europaea* (0.21) are lower than the values reported in the current study, suggesting fewer use reports in Morocco. Contrary to the present research, [33] discussed in their studies in the Chenab Riverine area of Punjab that the UV for *Ziziphus jujuba* is 0.60. Research work conducted in District Buner reported high use values of *Albizia lebbeck* (0.54) compared to our current research, indicating more use reports in the Buner district than in our study area. In Jammu and Kashmir, reported use values of *Butea variegata* (0.56) and *Ziziphus jujuba* (0.85). While the UV of *Capparis decidua* is the same as in our current study, the UV of other plants is comparatively higher in Jammu and Kashmir. The extraction of leaves or dried powder of *Albizia lebbeck* is utilized to cure diarrhea, tuberculosis, sore throats, skin

Table 3. Use value (UV).

Plant Species	Family	UV
Albizia lebbeck (L.) Benth.	Fabaceae	0.8
Azadirachta indica A. Juss.	Meliaceae	0.6
Bauhinia veriegata L.	Fabaceae	0.8
Cordia myxa L.	Boraginaceae	0.6
Dalbergia sissoo DC.	Fabaceae	0.6
Eucalyptus globulus Labill.	Myrtaceae	0.86
Ficus religiosa L.	Moraceae	0.6
Fraxinus ornus L.	Oleaceae	0.6
Laurus nobilis L.	Rubeaceae	0.6
Melia azedarach L.	Fabaceae	0.6
Moringa oleifera Lam.	Moringaceae	0.6
Morus alba L.	Moraceae	0.6
Olea europaea L.	Oleaceae	0.6
Phoenix dactylefera L.	Arecaceae	0.66
Prosopis cineraria (L.) Druce	Fabaceae	0.6
Senegalia catechu (L.f.) P.J.H. Hurter & Mabb.	Fabaceae	0.6
Tamarix aphylla (L.) H.Karst.	Tamaricaceae	0.6
Terminalia arjuna (Raxb. ex DC.) Wight & Arn.	Combreaceae	0.6
Vachellia nilotica (L.) P.J.H. Hurter & Mabb.	Fabaceae	0.8
Zizipus jujube Mill.	Rhamnaceae	0.8

https://doi.org/10.1371/journal.pone.0328913.t003

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diseases, and wounds. *Azadirachta indica* leaves are used as an extract or in boiled form to cure cough, fever, liver diseases, gastrointestinal discomfort, and rheumatism. Previously reported different plants were used to cure gastrointestinal discomfort, urinary tract infections, respiratory disorders, and cancer. The fruit and leaves of *Butea variegata* are employed to treat fever, diarrhea, diabetes, and ulcers. *Crataegus myxa* leaves are used in decoction form to cure cough, fever, and rheumatism. Terminalia arjuna is utilized to treat jaundice, skin diseases, and wounds.

Fidelity level (FL)

The fidelity level is a statistic for identifying plant species that local residents prefer to employ to treat a specific disease, in comparison to other plants used for the same reason. This index helps specify the most favored species cited by indigenous people for curing ailments (Table 4). In the present study, FL values were determined ranging between 68% and 33%. Higher FL values indicate extensive use of a particular plant in the area for treating ailments compared to those with lower FL values. *Eucalyptus globulus* achieved the highest FL (86%) for cough, highlighting the prevalence of this ailment in Hasilpur. *Albizia lebbeck*, *Azadirachta indica*, *Melia azedarach*, *Moringa oleifera*, *Olea europaea* and *Prosopis cineraria* shared an FL value of 60% for stomach diseases, while *Crataegus myxa, Dalbergia sissoo, Morus alba* and *Tamarix aphylla* also had an FL of 60% for liver diseases. *Butea monosperma* (80%) and *Laurus nobilis* (60%) exhibited high FL values for skin diseases, *Phoenix dactylifera* (66%) for stomach diseases, and *Vachellia nilotica* (80%). *Ficus religiosa, Fraxinus ornus, and Acacia catechu* held an FL value of 60% for diarrhea, while *Terminalia arjuna* and *Ziziphus jujuba* showed an FL of 60% for anti-inflammation. Plant species with high FL values are of particular interest in biological, phytochemical, and pharmacological studies to assess and validate their potential as new treatments and plant products.

The selected plant species with high FL values emerge as intriguing candidates for additional biological, phytochemical, and pharmacological studies seeking to evaluate their therapeutic potential and investigate the development of novel

Table 4. Fidelity level (FL %) values of plant species based on ethnomedicinal use.

Plant Species	Family	Diseases treated	FL %
Albizia lebbeck (L.) Benth.	Fabaceae	Stomach diseases	60
Azadirachta indica A. Juss.	Meliaceae	Stomach diseases	60
Bauhinia veriegata L.	Fabaceae	Skin diseases	80
Cordia myxa L.	Boraginaceae	Liver diseases	60
Dalbergia sissoo DC.	Fabaceae	Liver diseases	60
Eucalyptus globulus Labill.	Myrtaceae	Skin diseases	86
Ficus religiosa L.	Moraceae	Diarrhea	60
Fraxinus ornus L.	Oleaceae	Diarrhea	66
Laurus nobilis L.	Rubeaceae	Skin diseases	60
Melia azedarach L.	Fabaceae	Stomach diseases	60
Moringa oleifera Lam.	Moringaceae	Stomach diseases	60
Morus alba L.	Moraceae	Liver diseases	60
Olea europaea L.	Oleaceae	Stomach diseases	60
Phoenix dactylefera L.	Areaceae	Stomach diseases	60
Prosopis cineraria (L.) Druce	Fabaceae	Stomach diseases	60
Senegalia catechu (L.f.) P.J.H. Hurter & Mabb.	Fabaceae	Stomach diseases	60
Tamarix aphylla (L.) H. Karst.	Tamaricaceae	Liver diseases	60
Terminalia arjuna (Raxb. ex DC.) Wight & Arn.	Combreaceae	Anti-inflammatory	60
Vachellia nilotica (L.) P.J.H. Hurter & Mabb.	Fabaceae	Liver diseases	80
Ziziphus jujube Mill.	Rhamnaceae	Anti-inflammatory	60

https://doi.org/10.1371/journal.pone.0328913.t004

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medications and plant-derived products. This study provides unique insights into the ethnomedicinal practices of the Hasilpur region, emphasizing certain plant species vital for addressing frequent health concerns among the local residents. In contrast to the findings in Wazirabad, where *Eucalyptus globulus* showed an FL of 42.8% for flu and *Melia azedarach* exhibited 87.5% for measles, our study in Hasilpur indicates variations in the preferred plant species for these ailments. Similarly, when compared with the research conducted in District Karnataka, India, where *Justicia adhatoda* had an FL value of 100 for curing bronchitis, the FL values in our study suggest distinct plant preferences in Hasilpur. Additionally, in Madhupur, Bangladesh, *Terminalia chebula* recorded an FL value of 100 for treating gastrointestinal disorders, demonstrating a different pattern than observed in Hasilpur [34]. The low-fidelity proportion in our study implies that certain medicinal herbs are utilized less frequently by the local population. These components delineate the fidelity percentage of plants in the study area, showing their utilization and benefits by the common people. Variations were noted when comparing our results with studies conducted in Kurram Agency, Punjab, and Jammu and Kashmir [35]. For instance, *O. europaea* in Kurram Agency exhibited an FL of 75% for skin diseases, higher than our current study, where *O. europaea* was reported to treat a broader spectrum of diseases. Similarly, *Z. jujuba* in Jammu and Kashmir showed an FL percentage of 98 for diabetes, surpassing the FL percentage observed in the current study for *Z. jujuba*. These comparative analyses highlight the contextual variations in plant usage for specific ailments across diverse regions.

Comparison of ethnomedicinal data - Jaccard Index (JI)

The comparative examination of ethnomedicinal data acquired in our research region with findings from aligned regions reveals both differences and similarities, showing the significance of survey data in understanding traditional knowledge within the study area. Ethnomedicinal research is influenced by historical, organoleptic, and phytochemical factors, shaping the choices made by communities. To assess the novelty of our work, we scrutinized the collected data against previously published studies from various parts of the Punjab province. This comparative assessment aimed to evaluate similarities and differences at the regional and global levels. A total of 20 previously published articles from neighboring locations were considered for this analysis. The degree of resemblance between species recorded in the research region and those described in previous studies from surrounding locations was quantified using the Jaccard index, yielding insights into the comparative aspects of ethnomedicinal data across different geographical regions (Table 5). Jaccard index comparisons revealed varying degrees of similarity between the Yazman, Ahmadpur East, Bahawalnagar, and Bahawalpur regions in Punjab. The lowest JI was observed for the Yazman region, indicating a lack of commonality with the current study. In contrast, the highest JI was found for the Ahmadpur East area, suggesting a relatively higher degree of similarity in ethnomedicinal plant use. A lower JI value implies less comparability between the data from two regions, while a higher value signifies greater similarity. The presence of common species at both research sites may contribute to higher JI values. Notably, the current study reported new species or unique applications, contributing to lower JI values. Specific comparisons include a JI value of 6 with Bahawalnagar, indicating 4 common species with 3 having similar uses and 1 with dissimilar uses. Against Bahawalpur, the JI value is 4, with 5 common species, 3 with similar uses, and 2 with dissimilar uses. In comparison with Yazman, the JI value is zero due to no common species. Finally, when compared with Ahmadpur East, the JI value is 13, with 7 common species, 2 with similar uses, and 5 with dissimilar uses, emphasizing the diversity and uniqueness of ethnomedicinal knowledge across different Tehsils in Punjab.

Medicinal properties of selected plant species

Albizia lebbeck (L.) Benth.: *A. lebbeck*, commonly known as siris, holds a significant place in traditional Ayurvedic medicine owing to its anti-inflammatory and antimicrobial properties. Studies have identified various active constituents in *A. lebbeck*, including flavonoids, saponins, and alkaloids, which contribute to its therapeutic effects. Research performed by [36] demonstrated the potent anti-inflammatory activity of *A. lebbeck* bark extract in animal models of inflammation.

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Table 5. Jaccard index (JI) based on ethnomedicinal use.

Area	SY	TRSs	NPSU	NPDU	TSCBA	SEOSA	PPSU	PPDU	JI	Citation
Ahmad Pur East	2015	41	2	5	7	20	28.9	71.4	13	[55]
Vehari	2013	77	4	2	6	20	66.7	33.3	7	[21]
Bahawalnagar	2011	56	3	1	4	20	25	75	6	[57]
Kabirwala	2021	58	2	2	4	20	50	50	6	[22]
Rajanpur	2014	18	1	1	2	20	50	50	6	[58]
Dera Ghazi Khan	2019	189	4	5	9	20	44.4	55.6	5	[59]
Okara	2019	126	3	4	7	20	42.9	57.1	5	[1]
Sahiwal	2008	41	2	1	3	20	66.6	33.3	5	[60]
Bahawalpur	2022	111	3	2	5	20	66.6	34.4	4	[35]
Lodhran	2010	89	2	2	4	20	50	50	4	[61]
Gujrat	2019	88	2	2	4	20	50	50	4	[62]
Sheikhupura	2020	88	2	2	4	20	50	50	4	[55]
Gojra	2020	100	2	1	3	20	66.6	33.3	3	[29]
Sialkot	2011	48	1	1	2	20	50	50	3	[23]
Layyah	2020	106	2	2	4	20	50	50	3	[53]
Wazirabad	2014	85	1	1	2	20	25	20	2	[26]
Chakwal	2012	29	1	0	1	20	100	0	2	[1]
Gujranwala	2011	71	1	1	2	20	50	50	2	[43]
Yazman Mandi	2019	118	0	0	0	20	0	0	0	[54]
Toba Tek Singh	2022	84	0	0	0	20	0	0	0	[63]
Jatoi	2015	45	0	0	0	20	0	0	0	[55]
Nigeria	2003	45	0	0	0	20	0	0	0	[64]
India	2022	66	0	0	0	20	0	0	0	[65]

key: SY. study area, TRSS. The total number of reported species, NPSU. The number of plants with similar uses, NPDU. The number of plants with dissimilar uses, TSCBA. The total number of species common in both areas, PPSU. Percentage of plants with similar uses, PPDU. Percentage of plants with dissimilar uses, JI. Jaccard index

https://doi.org/10.1371/journal.pone.0328913.t005

Moreover, comparisons with other studies conducted in similar regions have consistently supported its traditional use as an anti-inflammatory agent, validating its medicinal efficacy.

Azadirachta indica A. Juss.: *A. indica*, commonly known as neem, is renowned for its broad spectrum of medicinal properties, including antibacterial, antiviral, and antifungal activities. Key active constituents found in neem include azadirachtin, nimbin, and quercetin. [37] conducted research highlighting the potent antimicrobial activity of neem leaf extracts against various pathogens. This finding aligns with numerous studies conducted globally, affirming the antimicrobial efficacy of neem and reinforcing its extensive use in traditional medicine across different regions.

Bauhinia veriegata L.: *B. variegata*, commonly known as kachnar, is known as a traditional medicinal plant due to its antioxidant and anti-diabetic properties. The plant contains various active constituents, including flavonoids, phenolic acids, and tannins. [38] conducted a study demonstrating the anti-diabetic effects of *B. variegata* leaf extract in diabetic rats, suggesting its potential therapeutic use in managing diabetes. Consistent findings from similar studies conducted in the region further validate its anti-diabetic potential.

Cordia myxa L.: *C. myxa*, commonly known as sebesten plum, has been traditionally used for its medicinal properties, including its role in managing respiratory disorders and promoting digestive health. Active ingredients, including flavonoids and tannins, contribute to its medicinal effects. While pharmacological studies specifically on *C. myxa* are limited,

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its traditional use in managing respiratory and digestive ailments has been documented in various ethnobotanical surveys conducted in the region, indicating its significance in traditional medicine.

Dalbergia sissoo DC.: *D. sissoo*, also known as Indian rosewood, is widely used as medicine due to its numerous therapeutic characteristics. The plant contains various active constituents, including flavonoids, terpenoids, and phenolic compounds, which contribute to its therapeutic effects. While specific pharmacological studies on Dalbergia sissoo are limited, ethnobotanical surveys and traditional knowledge indicate its use in managing various ailments such as inflammation, pain, and gastrointestinal disorders [39].

Eucalyptus globulus Labill.: *E. globulus*, commonly known as Tasmanian blue gum, is known for its medicinal properties, particularly its role in respiratory health and wound healing. The plant contains essential oils rich in compounds such as eucalyptol, which exhibit antimicrobial and anti-inflammatory activities. Pharmacological studies have demonstrated the efficacy of eucalyptus essential oil in relieving respiratory symptoms and promoting wound healing. Additionally, traditional uses of eucalyptus in managing respiratory ailments have been documented in various cultures, aligning with its pharmacological properties [40].

Ficus religiosa L.: *F. religiosa*, commonly known as the sacred fig or peepal tree, holds cultural and medicinal significance in various traditional systems of medicine. The plant includes bioactive components such as flavonoids, tannins, and alkaloids, which contribute to the therapeutic effects. Pharmacological studies have explored its diverse therapeutic effects, including anti-inflammatory, analgesic, and antimicrobial activities. Furthermore, ethnobotanical surveys have documented its traditional uses in managing gastrointestinal disorders, skin ailments, and respiratory conditions, highlighting its importance in traditional medicine systems.

Fraxinus ornus L.: *F. ornus*, commonly known as manna ash, possesses medicinal properties traditionally used for its anti-inflammatory and diuretic effects. Active constituents such as flavonoids and phenolic acids contribute to its therapeutic properties [41]. While specific pharmacological studies on *F. ornus* are limited, its traditional use in managing inflammatory conditions and promoting diuresis has been documented in ethnobotanical literature and traditional medicine practices.

Laurus nobilis L.: *L. nobilis* known as bay laurel or bay leaf, is valued for its aromatic and medicinal properties. The plant contains essential oils rich in compounds such as eucalyptol and cineole, which exhibit antimicrobial and anti-inflammatory effects. Pharmacological studies show the therapeutic potential of bay leaf in managing digestive disorders, respiratory ailments, and skin conditions [42]. Additionally, its traditional uses in various cuisines and folk medicine practices further highlight its cultural and medicinal significance.

Melia azedarach L.: *M. azedarach*, commonly known as dhrek, is renowned for its diverse medicinal properties traditionally used in Ayurvedic medicine. The plant contains bioactive compounds such as azadirachtin, nimbin, and quercetin, which exhibit antibacterial, antiviral, and antifungal activities. Pharmacological studies have validated its traditional uses, demonstrating its efficacy in managing skin disorders, oral health issues, and inflammatory conditions [37]. Furthermore, ethnobotanical surveys have documented its traditional uses in various cultures, highlighting its importance in traditional medicine systems.

Moringa oleifera Lam.: *M. oleifera* is known as the drumstick tree or moringa, is esteemed for its nutritional and medicinal properties. The plant is high in bioactive compounds such as vitamins, minerals, and antioxidants, which contribute to its therapeutic effects. Pharmacological studies have elucidated its diverse health benefits, including anti-inflammatory, antioxidant, and hypoglycemic activities. Traditional uses of moringa in managing various ailments such as malnutrition, diabetes, and inflammation are well-documented across different cultures, underscoring its importance in traditional medicine practices [43].

Morus alba L.: *M. alba*, commonly known as white mulberry, has been used traditionally for its medicinal properties, particularly in managing metabolic disorders such as diabetes. The plant includes bioactive substances such as flavonoids, alkaloids, and phenolic acids, which contribute to the medicinal effects. Pharmacological studies have

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demonstrated the anti-diabetic, hypolipidemic, and antioxidant activities of *M. alba* extracts. Additionally, ethnobotanical surveys have documented its traditional uses in various cultures, further validating its medicinal efficacy [44].

Olea europaea L.: *O. europaea*, commonly known as the olive tree, holds significant medicinal and cultural importance. The plant contains bioactive chemicals such as oleuropein, hydroxytyrosol, and flavonoids, which exhibit antioxidant, anti-inflammatory, and cardioprotective effects. Pharmacological research has validated the traditional applications, demonstrating its efficacy in managing cardiovascular disorders, inflammatory conditions, and metabolic syndrome [45]. Moreover, the olive tree holds cultural significance in Mediterranean cuisine and traditional medicine practices, symbolizing health and well-being.

Phoenix dactylifera L.: *P. dactylifera*, commonly known as the date palm, is revered for its medicinal properties. The date palm fruit contains important nutrients, vitamins, and minerals, making it an excellent food source in arid environments. Pharmacological studies have highlighted its diverse health benefits, including antioxidant, anti-inflammatory, and hepatoprotective activities. Traditional uses of dates in promoting digestive health, boosting energy, and supporting overall well-being are well-documented across various cultures, underscoring its significance in traditional medicine and culinary practices [46].

Prosopis cineraria (L.) Druce: *P. cineraria*, commonly known as khejri, holds cultural and medicinal importance in arid regions. The plant is drought-resistant properties and medicinal uses in traditional medicine systems. Flavonoids, alkaloids, and tannins are active elements that contribute to the medicinal effects. While specific pharmacological studies on *P. cineraria* are limited, ethnobotanical surveys and traditional knowledge indicate its use in addressing various disorders such as gastrointestinal disorders, respiratory ailments, and skin conditions [47].

Senegalia catechu (L.f.) P.J.H. Hurter & Mabb.: *S. catechu*, commonly known as katha or acacia catechu, is valued for its astringent and medicinal properties. The heartwood of the tree includes beneficial components like flavonoids and catechins, which exhibit antioxidant, anti-inflammatory, and antimicrobial activities. Pharmacological studies have demonstrated its efficacy in oral health, wound healing, and gastrointestinal disorders [48]. Additionally, traditional uses of katha in traditional medicine practices such as Ayurveda and traditional Chinese medicine further highlight its medicinal significance.

Tamarix aphylla (L.) H. Karst.: *T. aphylla*, commonly known as athel pine, possesses medicinal properties traditionally used for its astringent and anti-inflammatory effects. Active constituents such as tannins and flavonoids contribute to its therapeutic properties. While specific pharmacological studies on *Tamarix aphylla* are limited, its traditional use in managing inflammatory conditions and promoting wound healing has been documented in ethnobotanical literature and traditional medicine practices [49].

Terminalia arjuna (Roxb. ex DC.) Wight & Arn.: *T. arjuna*, commonly known as arjuna, holds a prominent place in Ayurvedic medicine for its cardioprotective properties. The bark of the arjuna tree includes bioactive substances such as flavonoids, tannins, and triterpenoids that have antioxidant, anti-inflammatory, and cardioprotective properties. Pharmacological studies have demonstrated its efficacy in managing cardiovascular disorders such as hypertension, arrhythmias, and heart failure. Additionally, traditional uses of arjuna in supporting heart health and managing cardiac ailments are well-documented in Ayurvedic texts and traditional medicine practices [50].

Vachellia nilotica (L.) P.J.H. Hurter & Mabb.: *V. nilotica*, commonly known as Babul is well-known for its medicinal properties, which have long been utilized in Ayurvedic and traditional medicine systems. The bark and pods of the *Acacia nilotica* tree contain bioactive substances like tannins, alkaloids, and flavonoids, which have antibacterial, anti-inflammatory, and analgesic properties. Pharmacological studies have demonstrated its efficacy in managing oral health, gastrointestinal disorders, and skin conditions [51].

Ziziphus jujuba Mill.: *Z. jujuba*, commonly known as jujube or Chinese date, is valued for its medicinal and nutritional properties. The fruit of the jujube tree is rich in vitamins, minerals, and antioxidants, making it a valuable food source with potential health benefits. Pharmacological studies have highlighted its diverse therapeutic effects, including antioxidant,

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anti-inflammatory, and neuroprotective activities. Additionally, traditional uses of jujube in promoting digestive health, enhancing sleep quality, and boosting immunity are well-documented in traditional medicine practices across different cultures [52].

Novelty index

The current study investigated both the study area and affiliated regions. The selection of research articles for review was conducted randomly. From 2006 to 2022, studies published in Pakistan were studied, while three research publications from associated countries were evaluated from 2011 to 2021. This is the first ethnobotanical research in Tehsil Hasilpur, and we discovered seven distinct ethnobotanical applications of plants that have not been recorded in previous studies. It was discovered that the bulk of the plants mentioned in this study are restricted to their current geographical location. Some plants from this research region were extensively dispersed in linked locations because of their broad adaptation in diverse ecological zones and were reported for medical applications; however, during the comparison investigation, some plants with novel medicinal uses were discovered. These include *S. catechu, V. nilotica, F. religosa, F. ornus, P. cineraria* and *T. aphylla.* These plants were found to treat diseases that were not previously cited in any other research findings. (Table 6). The medicinal applications of *A. lebbeck* involve extracting leaves or dried powder to address ailments such as diarrhea, cancer and inflammation. These uses align with prior reports [53] that highlighted the plant's efficacy in treating asthma and blood purification diseases. *A. indica*, known for its diverse healing properties, sees its leaves utilized as an extract or in boiled form to alleviate stomach issues, skin conditions, and inflammation. Earlier research [54] also emphasized its role in addressing gastrointestinal discomfort, urinary tract infections, respiratory disorders, and cancer. *B. variegata*, valued for its therapeutic attributes, employs both fruit and leaves in treating fever, diarrhea, diabetes, and ulcers, in line with previous findings [55] highlighting its sedative, diuretic, and hypnotic properties. *C. myxa* decoction form is employed to remedy cough, fever, and rheumatism, complementing past uses [56] for malaria fever and rheumatism. The leaves of *D. sissoo*, prepared as a decoction, address rheumatism, cough, and heart disease,

M. oleifera's medicinal scope includes cancer, heart, and skin diseases, and ulcers, echoing prior research [67,68] that explored its efficacy in treating sore eyes, boils, and swollen hands and feet. *P. cineraria* tackle fever, diarrhea, and heart diseases, administered as a decoction, aligning with previous uses [69] for antispasmodic effects in asthma. *Sanagelia catechu* finds application in treating skin diseases and stomachaches, through past uses [70,71] for liver and spleen issues. *T. arjuna* addresses jaundice, skin diseases, and wounds, while *O. europea* tackles cancer and gastrointestinal discomfort, mirroring earlier applications [72] for diarrhea, respiratory, and urinary tract infections. *V. nilotica*, recognized for its healing properties is employed to address tuberculosis, stomachaches, and hepatitis, aligning with past uses [73] for treating skin diseases.

Conclusion

Our study shows intricate interplay of socio-demographic factors in shaping traditional medicinal practices within Tehsil Hasilpur, Punjab. Through a thorough examination of 300 respondents, we discover significant gender imbalances, educational gradients, and occupational influences that delineate the transmission and preservation of indigenous healthcare knowledge. The pronounced gender gap, with males comprising 85.6% of respondents, emphasizes cultural norms dictating access to outdoor environments and knowledge acquisition. Notably, traditional healing wisdom transcends formal education, flourishing among individuals with primary education (53%) and even the illiterate (24%), emphasizing the role of informal knowledge dissemination. The prevalence of perennial plants (98.6%) adapted to the arid climate and the predominance of leaves as the primary plant part used in remedies (65.33%), primarily prepared as decoctions (24%), signify a nuanced understanding of local flora and preparation techniques. High Informant Consensus Factor (ICF) values for specific ailments and notable Use Values (UV) for plants like *Eucalyptus globulus* highlight the efficacy and importance

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Table 6. Novelty index and ethnobotanical profile of plants from study area.

BN/VN	LN	Family	Life form	Habitat	PU	Rec	App	Uses
Senegalia catechu (L.f.) P.J.H. Hurter & Mabb. IUB/BOT/23/40091	Khaar	Fabaceae	Perennial	Tree	Leaves	Powder	Oral	Used to treat cancer, diarrhea, and anti-inflammatory.
Leucaena leucocephala (Lam.) de Wit. IUB/BOT/23/40092	Jumbey		Perennial	Tree	Leaves	Powder	Oral	Stomach diseases, anti-inflammatory, and skin diseases
Albizia lebbeck (L.) Benth. IUB/BOT/23/40093	Sheren		Perennial	Tree	Leaves	Extraction	Oral	Stomach diseases, anti-inflammatory, and skin diseases
Bauhinia variegata L. IUB/BOT/23/40094	Kachnar		Perennial	Tree	Leaves	juice	Oral	Wound healing, anti-inflammatory, and skin diseases
Vachellia nilotica (L.) P.J.H.Hurter & Mabb. IUB/BOT/23/40095	Phari kikar		Perennial	Tree	Leaves	powder	Oral	Liver disease, stomach diseases, and teeth treatment
Morus alba L. IUB/BOT/23/40096	Shahtoot	Moraceae	Perennial	Tree	Leaves Fruits	powder	Oral	Stomach diseases, anti-inflammatory, and Liver diseases
Ficus religiosa L. IUB/BOT/23/40097	Pipel		Perennial	Tree	Leaves Fruits	powder	Oral	Diarrhea, anti-inflammatory, and skin diseases
Olea europia L. IUB/BOT/23/40098	Zaitoon	Oleaceae	Perennial	Tree	Leaves		Oral	Wound healing, anti-inflammatory, and stomach diseases
Fraxinus ornus L. IUB/BOT/23/40099	Ash plant		Annual	Tree	Leaves Fruits	powder	Oral	Diahorea, anti-inflammatory, and skin diseases
Azadirachta indica A. Juss. IUB/BOT/23/40100	Neem	Meliaceae	Perennial	Tree	Leaves Fruits	juice	Oral	Stomach diseases, anti-inflammatory, and skin diseases
Laurus nobilis L. IUB/BOT/23/40101	Rind	Rubiaceae	Perennial	Tree	Leaves Fruits	powder	Oral	Stomach diseases, anti-inflammatory, and skin diseases
Moringa oleifera Lam. IUB/BOT/23/40102	Suhanjna	Moringaceae	Perennial	Tree	Leaves	powder	Oral	Stomach diseases, anti-inflammatory, and skin diseases
Ziziphus jujuba Mill. IUB/BOT/23/40103	Bere	Rhamnaceae	Perennial	Tree	Leaves powder	juice	Oral	Anti-inflammatory and skin diseases
Terminalia arjuna (Raxb. ex DC.) Wight & Arn. IUB/BOT/23/40104	Arjun	Combretaceae	Perennial	Tree	Leaves	powder	Oral	Wound healing, anti-inflammatory, and stomach diseases
Phoenix dactlylfera L. IUB/BOT/23/40105	Khajoor	Arecaceae	Perennial	Tree	Leaves powder	juice	Oral	Stomach diseases, anti-inflammatory, and skin diseases
Prosopis cineraria (L.) Druce IUB/BOT/23/40106	Jhund	Fabaceae	Perennial	Tree	Leaves	juice	Oral	Stomach diseases, anti-inflammatory, and skin diseases
Dalbergia sissoo DC. IUB/BOT/23/40107	Tali	Fabaceae	Perennial	Tree	Leaves	powder	Oral	Wound healing, anti-inflammatory, and liver diseases
Cordia myxa L. IUB/BOT/23/40108	Lasora	Boraginaceae	Perennial	Tree	Leaves	juice	Oral	Liver disease, laxative, and anti-inflammatory

(Continued)

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Table 6. (Continued)

BN/VN	LN	Family	Life form	Habitat	PU	Rec	App	Uses
Tamarix aphylla (L.) H. Karst. IUB/BOT/23/40109	Farash	Tamarica-ceae	Perennial	Tree	Leaves	powder	Oral	Liver disease, wound healing, and anti-inflammatory
Eucalyptus globulus Labill. IUB/BOT/23/40110	Sfeda	Myrtaceae	Perennial	Tree	Leaves Fruits	powder	Oral	Teeth treatment, fever, and skin diseases

Abbreviations: BN. Botanical Name; VN. Voucher number. LN. Local name, LF. Life form, PU. Part used Rec. Recipe, App. Mode of application, (UV). Use value, N. Number of informants,

*Novel uses. *Senegalia catechu*: Excessive ingestion may cause nausea, vomiting, and in some cases, allergic reactions; *Melia Azadirach:* Ingesting neem oil or large quantities of neem leaves may lead to nausea, vomiting, and diarrhea; *Azadirachta indica*: Neem oil can be toxic if ingested in large amounts, causing nausea, vomiting; *Phoenix dactylifera*: Excessive consumption of date pits might lead to intestinal blockage; *Eucalyptus globulus*: Ingesting eucalyptus oil can be toxic and may cause nausea, vomiting and depression consistent with earlier applications [61–66] for pain and swelling, deafness, wound healing, and skin diseases.

https://doi.org/10.1371/journal.pone.0328913.t006

of traditional remedies in addressing prevalent health concerns. These findings emphasize the resilience of traditional healing practices, suggesting their integration into mainstream healthcare policies for sustainable health outcomes. Additionally, our study identifies novel medicinal uses of plant species, indicating avenues for further pharmacological investigations and collaborative efforts between local communities, researchers, and policymakers should guarantee the regional ethnomedicinal resources are conserved and used sustainably.

Author contributions

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